

subgraph of G is the intersection graph of a compatible collection of subtrees of T . Thus, by Theorem 3.2 it suffices to show that G has a simple vertex.

Let $V(G) = \{v_1, \dots, v_n\}$ and let $T_1, \dots, T_n$ be subtrees of T in Λ satisfying $v_i \sim v_j$ in G if and only if $V(T_i) \cap V(T_j) \neq \emptyset$. For each i , let w_i be the unique vertex of T_i of minimum weighted distance to r . We may assume, without loss of generality, that $d_T(w_1, r) \geq d_T(w_j, r)$ for $j = 2, \dots, n$. We claim that v_1 is simple in G . Suppose that $v_i, v_j \in N_G(v_1)$ and $T_i > T_j$. It suffices to show that $N_G(v_j) \subseteq N_G(v_i)$. Since

$$d_T(w_1, r) \geq d_T(w_j, r)$$

and

$$V(T_1) \cap V(T_j) \neq \emptyset$$

it follows that

$$w_1 \in V(T_j).$$

Similarly

$$w_1 \in V(T_i).$$

Since $T_i > T_j$ we have

$$w_j \in V(T_i).$$

Let $v_k \in N_G(v_j)$. Then

$$V(T_j) \cap V(T_k) \neq \emptyset.$$

If $w_j \in V(T_k)$ then $V(T_i) \cap V(T_k) \neq \emptyset$. Otherwise

$$w_k \in V(T_j)$$

$$d_T(r, w_k) \leq d_T(r, w_1)$$

and

$$w_1 \in V(T_i) \cap V(T_j)$$

whence

$$w_k \in V(T_i)$$

since $T_i > T_j$. In either event

$$V(T_i) \cap V(T_k) \neq \emptyset$$

and so

$$v_k \in N_G(v_i).$$

Since v_k was an arbitrary vertex of $N_G(v_j)$, it follows that

$$N_G(v_j) \subseteq N_G(v_i)$$

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□